

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-  
a1ff6496db1f6e019.jsonl`  
- SHA-256: `4723f521bad2e73d69436e1b2dbe65773907bc8804bc0ba19c2df6a40543bf17`  
- Source modified: `2026-06-16T03:24:43+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:23:44.346Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"TriDet introduces a 'Trident-head' that models action boundaries via an estimated relative probability distribution around the boundary, addressing imprecise/ambiguous boundary predictions ? directly relevant to the over-merge problem of splitting and precisely locating start/end of rallies in a dense per-frame signal."

****Source:**** <https://arxiv.org/abs/2303.07347> (primary)

****Supporting quote:**** "we propose a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary. Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:23:48.395Z)

I'll verify this claim about TriDet's Trident-head. Let me start by examining the primary source.

3. assistant (2026-06-16T03:23:51.487Z)

I'll fetch the primary arXiv source to verify the claim about TriDet's Trident-head.

4. user (2026-06-16T03:23:59.761Z)

TriDet's Trident-Head and Action Boundary Modeling

Based on the abstract provided, here's what we know about TriDet's approach:

The Problem It Solves

TriDet addresses a fundamental challenge in temporal action detection: "Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos."

The Trident-Head Solution

The paper proposes "a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary." Rather than predicting exact frame-level boundaries, this component models the uncertainty and ambiguity inherent in action transitions by estimating probability distributions relative to the boundary location.

Performance Gains

The framework achieves significant improvements: TriDet "achieves state-of-the-art performance on three challenging benchmarks: THUMOS14, HACS and EPIC-KITCHEN 100, with lower computational costs." Specifically on THUMOS14, it reached 69.3% average mAP while outperforming previous best methods by 2.5% with only 74.6% of the computational latency required.

The exact mechanism of how the Trident-head functions internally is not detailed in the abstract; that information would require examining the full paper.

5. user (2026-06-16T03:24:03.181Z)

Web search results for query: "TriDet Trident-head action boundary relative probability distribution temporal action detection CVPR 2023"

Links: [{"title": "TriDet - CVPR 2023 Open Access Repository", "url": "https://openaccess.thecvf.com/content/CVPR2023/html/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.html"}, {"title": "TriDet: Temporal Action Detection", "url": "https://www.emergentmind.com/papers/2303.07347"}, {"title": "CVPR Poster TriDet: Temporal Action Detection With Relative Boundary Modeling", "url": "https://cvpr.thecvf.com/virtual/2023/poster/21956"}, {"title": "[2303.07347] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/abs/2303.07347"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/10203543/"}, {"title": "[Quick Review] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://liner.com/review/tridet-temporal-action-detection-with-relative-boundary-modeling"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling Dingfeng Shi?", "url": "https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/pdf/2303.07347"}, {"title": "TriDet/README.md at master · dingfengshi/TriDet", "url": "https://github.com/dingfengshi/TriDet/blob/master/README.md"}, {"title": "GitHub - dingfengshi/TriDet: [CVPR2023] Code for the paper, TriDet: Temporal Action Detection with Relative Boundary Modeling · GitHub", "url": "https://github.com/dingfengshi/TriDet"}]

I found comprehensive information about TriDet from CVPR 2023. Here's a summary of the key

findings:

Overview

TriDet is a one-stage framework for temporal action detection. Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos. To alleviate this problem, the paper proposes a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary.

Key Components

****Trident-head:**** The Trident-head is designed to precisely locate action boundaries based on relative boundary modeling by considering the relation of features in a certain period and obtaining the relative probability of being a boundary for each instant in that period. The Trident-head consists of three components: a start head, an end head, and a center-offset head.

****Feature Pyramid:**** In the feature pyramid of TriDet, the paper proposes a Scalable-Granularity Perception (SGP) layer to aggregate information across different temporal granularities, which is much more efficient than recent transformer-based feature pyramids.

Performance

TriDet achieves state-of-the-art performance on three challenging benchmarks: THUMOS14, HACS and EPIC-KITCHEN 100, with lower computational costs. For example, TriDet hits an average mAP of 69.3% on THUMOS14, outperforming the previous best by 2.5%, but with only 74.6% of its latency.

The code is available on GitHub at the official repository for the project.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:24:07.094Z)

The claim is well-supported. Let me verify the specific assertion about its relevance to "dense per-frame signal" and over-merge boundary splitting by checking the paper's actual mechanism more closely, and check for any contradicting evidence.

7. user (2026-06-16T03:24:17.756Z)

Web search results for query: "TriDet relative boundary modeling Trident-head start end center-offset head limitations criticism"

Links: [{"title": "TriDet: Temporal Action Detection", "url": "https://www.emergentmind.com/papers/2303.07347"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/pdf/2303.07347"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling Dingfeng Shi?", "url": "https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf"}, {"title": "[2303.07347] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/abs/2303.07347"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling | alphaXiv", "url": "https://www.alphaxiv.org/abs/2303.07347"}, {"title": "CVPR Poster TriDet: Temporal Action Detection With Relative Boundary Modeling", "url": "https://cvpr.thecvf.com/virtual/2023/poster/21956"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2023arXiv230307347S/abstract"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/10203543"}, {"title": "[Quick Review] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://liner.com/review/tridet-temporal-action-detection-with-relative-boundary-modeling"}, {"title": "Temporal Action Localization with Enhanced Instant Discriminability", "url": "https://arxiv.org/pdf/2309.05590"}]

Based on the search results, here's information about TriDet's relative boundary modeling and Trident-head:

Overview

TriDet is a one-stage framework for temporal action detection that addresses the problem of imprecise boundary predictions in untrimmed videos.

Trident-head and Relative Boundary Modeling

The Trident-head consists of three components: a start head, an end head, and a center-offset head, which are designed to locate the start boundary, end boundary, and the temporal center of the action, respectively.

Unlike methods that rely on segment-level global features or instant-level regressions, the Trident-head models action boundaries via a relative probability distribution approach, using start, end, and center-offset heads that estimate relative probabilities across adjacent time bins.

The predicted boundary response in neighboring bin set is summed element-wise with the center offset corresponding to each instant, which is further estimated as the relative boundary distribution, and finally the offset is computed based on the expected value of the bin.

Identified Limitations

The paper identifies two limitations: the rank loss problem and high computational complexity. To address the rank loss problem, the SGP (Scalable-Granularity Perception) layer was proposed to mitigate the rank loss problem of self-attention that occurs in video features.

Performance

TriDet achieves state-of-the-art performance on three challenging benchmarks (THUMOS14, HACS and EPIC-KITCHEN 100) with lower computational costs, hitting an average mAP of 69.3% on THUMOS14, outperforming the previous best by 2.5%, but with only 74.6% of its latency.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:24:31.327Z)

I have enough to adjudicate. Let me assess the claim against the checklist.

****Claim accuracy assessment:****

1. ****Supported by quote?**** Yes. The abstract verbatim states: "we propose a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary" and "Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos." The claim restates this accurately ? Trident-head models boundaries via a relative probability distribution to address imprecise/ambiguous boundaries.

2. ****Source quality:**** Primary source is arXiv 2303.07347, which is a CVPR 2023 peer-reviewed paper ("TriDet: Temporal Action Detection with Relative Boundary Modeling"), with official code at github.com/dingfengshi/TriDet. SOTA on THUMOS14, HACS, EPIC-KITCHENS-100. Strong, primary, peer-reviewed source matching the claim's strength.

3. ****Current?**** 2023, well within the 2022?2025 window. Not outdated; TriDet remains a recognized TAL baseline.

4. ****Not marketing/forum speculation.**** Peer-reviewed conference paper.

****Scrutiny of the relevance / overreach portion:**** The claim adds an interpretive layer ? "directly relevant to the over-merge problem of splitting and precisely locating start/end of

rallies in a dense per-frame signal." This is the verifier's job to scrutinize:

- TriDet is genuinely a boundary-localization method (start/end/center heads), so "precisely locating start/end" is faithful.
- The "dense per-frame signal" framing is a reasonable application-level mapping: TAL operates on dense per-frame feature sequences, and the report dimension is explicitly about TAL methods for segmentation. This is application reasoning, not a factual claim about the paper.
- Minor nuance: TriDet is a full transformer-pyramid + anchor-free detector designed for hundreds of labeled videos (THUMOS14 ~200+ training videos), not directly the tiny-data 1-D scorer the system has. The claim does NOT assert TriDet runs on tiny data ? it asserts the *Trident-head idea* (relative boundary modeling) is relevant to splitting/locating boundaries. That is a defensible, correctly-scoped relevance statement, not an overreach. The interpretive "directly relevant" is a judgment the report author is entitled to make, and it maps correctly to the over-segmentation/boundary dimension.

The core factual claim is verbatim accurate from a strong primary peer-reviewed source, current, and the relevance framing is faithful rather than an overreach. I cannot refute it.

9. user (2026-06-16T03:24:38.088Z)

Structured output provided successfully

10. assistant (2026-06-16T03:24:43.150Z)

Verification complete. The claim is ****not refuted**** (confidence: high).

The core factual assertion is verbatim accurate from a strong primary source: arXiv 2303.07347 is the CVPR 2023 peer-reviewed paper "TriDet: Temporal Action Detection with Relative Boundary Modeling," with official code and SOTA results on THUMOS14/HACS/EPIC-KITCHENS-100. The supporting quote matches the abstract exactly, the mechanism (start/end/center-offset heads estimating relative boundary probability distributions) is confirmed across multiple sources, and the paper is current. The relevance framing to the over-merge/boundary-splitting problem is a faithful application mapping, not an overreach ? though I flagged one honest caveat that TriDet is a heavy transformer-pyramid detector validated on large benchmarks, so direct deployment on the tiny 1-D-trajectory setting is non-trivial (the claim only asserts the *idea* is relevant, which holds).